

Sapia / EduGap

Quiz-First Cost Report

Prepared for Mentor Review

Cohort: 500 enrolled users | Period: 3 months | Currency: USD

\$265.71 Expected 3-month cost	\$420 Recommended reserve	143,000 Quiz attempts	95.1% Low-case margin
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This report estimates software operating cost for a 500-user Sapia/EduGap cohort. The model reflects the intended use case: adaptive quiz is the primary workload, while AI chatbot is secondary and benefits from high cache reuse around slides, rubrics, concept summaries, and system prompts.

The report is an estimate for planning and mentor review. It should be replaced with measured COGS once production telemetry is available.

1. Executive Summary

Recommended estimate: **\$265.71** total software operating cost for 3 months using OpenAI gpt-5.4-nano as the default model strategy. This equals **\$0.53 per enrolled user**, **\$0.22 per active learner-month**, and **\$0.0019 per quiz attempt**.

Planning reserve: **\$420** for the 3-month cohort. The reserve includes roughly 40% buffer for retries, traffic spikes around deadlines, imperfect cache effectiveness during onboarding, and unmeasured chat usage.

If evidence-grade observability requires Braintrust Pro, use **\$1,150** as the practical reserve. This is not needed for the base case unless the mentor/customer requires longer trace retention, RBAC, team workflows, or richer evaluation reporting.

2. Cost Summary

Scenario	AI variable	Infra baseline	3-month total	Cost / user	Cost / quiz
Gemini 2.5 Flash-Lite	\$11.32	\$240.00	\$251.32	\$0.50	\$0.0018
OpenAI gpt-5.4-nano	\$25.71	\$240.00	\$265.71	\$0.53	\$0.0019
Gemini 2.5 Flash	\$51.04	\$240.00	\$291.04	\$0.58	\$0.0020
OpenAI gpt-5.4-mini	\$93.72	\$240.00	\$333.72	\$0.67	\$0.0023

3. Scope

Item	Scope
Product	Sapia / EduGap adaptive quiz and AI tutor product
Cohort size	500 enrolled users
Period	3 months / 13 weeks
Included	AI tokens, cached input, embeddings, core infra, cache, frontend hosting
Excluded	Human mentor labor, sales, tax, payment processing, legal, custom compliance, SSO

4. Key Assumptions

Assumptions are separated here because they are the main uncertainty. They should be replaced with actual telemetry after week 2 and month 1 of the first cohort.

Assumption	Value	Reason
Monthly active learners	400, 80% of enrolled	Cohort-led learning should outperform open MOOCs, but not every enrolled user will study every month.
Average active days/week	2.75	Quiz-first course workflow is a few active study days per week, not daily usage.
Average DAU	157 learners/day	400 MAU * 2.75 / 7. This implies 39.3% DAU/MAU, close to Duolingo Q4 2025 reported 39.6%.
Study time / active day	40 minutes	Equals 110 minutes/week, within the 1.5-2.5h/week online-course engagement band from IES REL.
Learner active-days	14,300	400 active learners * 2.75 days/week * 13 weeks.
Total platform time	9,533 hours	14,300 active-days * 40 minutes / 60.

5. Workload Model

Workload	3-month volume	Modeling reason
Quiz attempts	143,000	Main use case. 14,300 active-days * 10 quiz questions/day.
AI-assisted quiz calls	57,200	40% of attempts use AI for hints, explanation, short-answer review, or error coaching.
Chat sessions	2,145	Secondary use case. 0.15 chat sessions per learner active-day.
Mentor reports	13	One weekly cohort insight report.

Workload	3-month volume	Modeling reason
Evaluation samples	600	200/month for quiz explanation, citation, and guardrail checks.
Embedding volume	2.5M tokens	Course source bundle: slides, labs, docs, references.

6. Cache Model

The chatbot and quiz tutor are cache-heavy because requests repeat the same course context: slide summaries, concept rubrics, guardrail rules, system prompts, and question explanations.

Workload	Normal input	Cached input	Output
Quiz AI assistance	25,740,000	60,060,000	10,296,000
Chatbot	7,507,500	18,232,500	3,217,500
Mentor reports	325,000	715,000	65,000
Evaluations	960,000	840,000	150,000
Embeddings	2,500,000	0	0

7. AI Cost Breakdown

Workload	Gemini Flash-Lite	OpenAI nano	Gemini Flash	OpenAI mini
Quiz AI assistance	\$8.19	\$19.22	\$37.97	\$70.14
Chatbot	\$2.49	\$5.89	\$11.66	\$21.48
Mentor reports	\$0.08	\$0.16	\$0.31	\$0.59
Evaluations	\$0.18	\$0.40	\$0.73	\$1.46
Embeddings	\$0.38	\$0.05	\$0.38	\$0.05
Total AI cost	\$11.32	\$25.71	\$51.04	\$93.72

Interpretation: after cache, chatbot is not the main cost driver. On OpenAI gpt-5.4-nano, chatbot cost is only \$5.89 for the whole cohort. Quiz AI assistance is larger at \$19.22, but still small relative to infrastructure.

8. Infrastructure Baseline

Component	Monthly	3-month	Reason
Supabase Pro	\$25	\$75	Auth, Postgres, pgvector, RLS baseline.
Render backend Standard	\$25	\$75	More realistic than free service for 500-user cohort.
Render Key Value Starter	\$10	\$30	Redis/cache baseline.
Vercel Pro	\$20	\$60	Commercial frontend hosting assumption.
Braintrust Starter	\$0	\$0	Base case uses starter/free observability.
Total	\$80	\$240	Core operating baseline.

9. Revenue and Margin

Package	Formula	3-month revenue
Standard low	$(\$300 \text{ cohort} + \$3 * 500 \text{ learners}) * 3 \text{ months}$	\$5,400
Standard high	$(\$500 \text{ cohort} + \$8 * 500 \text{ learners}) * 3 \text{ months}$	\$13,500

Scenario	Total cost	Margin at \$5,400	Margin at \$13,500
Gemini 2.5 Flash-Lite	\$251.32	95.3%	98.1%

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OpenAI gpt-5.4-nano	\$265.71	95.1%	98.0%
Gemini 2.5 Flash	\$291.04	94.6%	97.8%
OpenAI gpt-5.4-mini	\$333.72	93.8%	97.5%

10. Recommended Budget Decision

Decision item	Recommended value
Expected base cost	\$265.71
Operational buffer	40%, or \$106.28
Rounded reserve for mentor/pilot planning	\$420
Evidence-grade reserve if Braintrust Pro is required	\$1,150

- Use \$420 as the default 3-month software COGS reserve for one 500-user cohort.
- Use \$1,150 only if paid evidence-grade observability is part of the mentor/customer promise.
- Keep the Standard pricing hypothesis unchanged; margin is strong under this usage model.
- Define fair-use limits for quiz attempts and chatbot sessions, but avoid overly tight limits because software COGS is low.

11. Calculation Audit

Core volume calculation:

Monthly active learners = $500 * 80\% = 400$
 Learner active-days = $400 * 2.75 * 13 = 14,300$
 Quiz attempts = $14,300 * 10 = 143,000$
 AI-assisted quiz calls = $143,000 * 40\% = 57,200$
 Chat sessions = $14,300 * 0.15 = 2,145$

OpenAI gpt-5.4-nano quiz cost:

normal input cost = $25,740,000 / 1,000,000 * 0.20 = 5.148$
 cached input cost = $60,060,000 / 1,000,000 * 0.02 = 1.2012$
 output cost = $10,296,000 / 1,000,000 * 1.25 = 12.87$
 quiz AI cost = $19.2192 = \$19.22$

OpenAI gpt-5.4-nano chatbot cost:

normal input cost = $7,507,500 / 1,000,000 * 0.20 = 1.5015$
 cached input cost = $18,232,500 / 1,000,000 * 0.02 = 0.36465$
 output cost = $3,217,500 / 1,000,000 * 1.25 = 4.021875$
 chatbot cost = $5.888025 = \$5.89$

12. References

Reference	Use
OpenAI API pricing - https://developers.openai.com/api/docs/pricing	OpenAI input, cached input, output prices.
Gemini API pricing - https://ai.google.dev/gemini-api/docs/pricing	Gemini model pricing and context/cache pricing.
Supabase pricing - https://supabase.com/pricing	Supabase Pro baseline.
Render pricing - https://render.com/pricing	Backend and cache baseline.
Vercel pricing - https://vercel.com/pricing	Frontend hosting baseline.
Braintrust pricing - https://www.braintrust.dev/pricing	Optional paid observability scenario.
Duolingo 2025 Form 10-K	DAU/MAU benchmark for learning habit products.
IES REL online course engagement study	Weekly online-course engagement benchmark.

13. Mentor Review Notes

- This is an estimate, not measured production COGS.
- The model intentionally treats quiz as the core workload and chatbot as secondary.
- The biggest validation need is telemetry: actual quiz attempts, chat sessions, cache hit rate, and token usage by feature.
- If actual cache hit is much lower than assumed, rerun the report after week 2 of pilot traffic.